

1 - INDEXES & LENGTH

Challenge 1: First & last

```
const fruits = ["apple", "banana", "cherry", "date"];
```

```
// 1. Log the first element  
// 2. Log the last element (use .length)  
// 3. Log the total count of fruits
```

Challenge 2: Update by index

```
const scores = [80, 55, 90, 42, 73];
```

```
// 1. Change the second score to 60  
// 2. Change the last score to 99  
// 3. Log the updated array
```

2 - LOOPS

Challenge 3: for loop - sum

```
const prices = [12, 5, 8, 30, 3];
```

```
// Calculate the total sum of all prices  
// using a for loop. Log the result.
```

Challenge 4: for...of - greet

```
const names = ["Alice", "Bob", "Charlie", "Diana"];
```

```
// Log "Hello, Alice!", "Hello, Bob!" ...  
// for each name using for...of
```

Challenge 5: forEach - multiply

```
const nums = [2, 4, 6, 8, 10];
```

```
// Log each number multiplied by 3  
// using .forEach()
```

3 - PUSH, POP, SHIFT, UNSHIFT

Challenge 6: Queue simulation

```
const queue = ["Tom", "Sara"];

// 1. Add "Mike" to the end
// 2. Add "Lena" to the beginning
// 3. Remove the first person (they got served)
// 4. Remove the last person (they left)
// 5. Log the final queue
```

4 - MAP

Challenge 7: Celsius to Fahrenheit

```
const temps = [0, 10, 20, 30, 40];

// Create a new array where each
// Celsius temp is converted to Fahrenheit.
// Formula: (C * 9/5) + 32
```

Challenge 8: Pass the threshold

```
const scores = [45, 82, 60, 37, 91, 55, 78];

// 1. Get only scores that are 60 or above
// 2. Get only scores below 60 (failing)
// Log both arrays
```

Challenge 9: Count occurrences

```
const votes = ["yes", "no", "yes", "yes", "no", "yes"];

// Use .reduce() to count how many times
// each answer appears.
// Expected result: { yes: 4, no: 2 }
```

7 - SORT & REVERSE

Challenge 10: Sort numbers

```
const nums = [40, 1, 5, 200, 13, 99];
```

```
// 1. Sort ascending (smallest first)  
// 2. Sort descending (largest first)  
// 3. Reverse the original array
```

8 - indexOf, lastIndexOf, includes, find, findIndex

Challenge 11: Search methods

```
const colors = ["red", "blue", "green", "blue", "red"];
```

```
// 1. What index is the first "blue"?  
// 2. What index is the last "blue"?  
// 3. Does it include "green"?  
// 4. Does it include "yellow"?
```

9 - SPLICE & SLICE

Challenge 12: slice - extract a copy

```
const days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"];
```

```
// 1. Get the first 3 days  
// 2. Get only the weekend (Sat, Sun)  
// 3. Log the original - confirm it's unchanged
```

Challenge 18: splice - mutate in place

```
const menu = ["soup", "salad", "pasta", "steak", "cake"];
```

```
// 1. Remove "pasta" (by index) using splice  
// 2. Insert "risotto" where "pasta" was  
// 3. Log the final menu
```